Altered brain activation during task-evoked back pain in chronic low back pain: An fMRI Study

*Vananderoye C, *Coppieters I, Weltens N, De Baets L, **Ashar YK, **Wager TD, **Van Oudenhove L

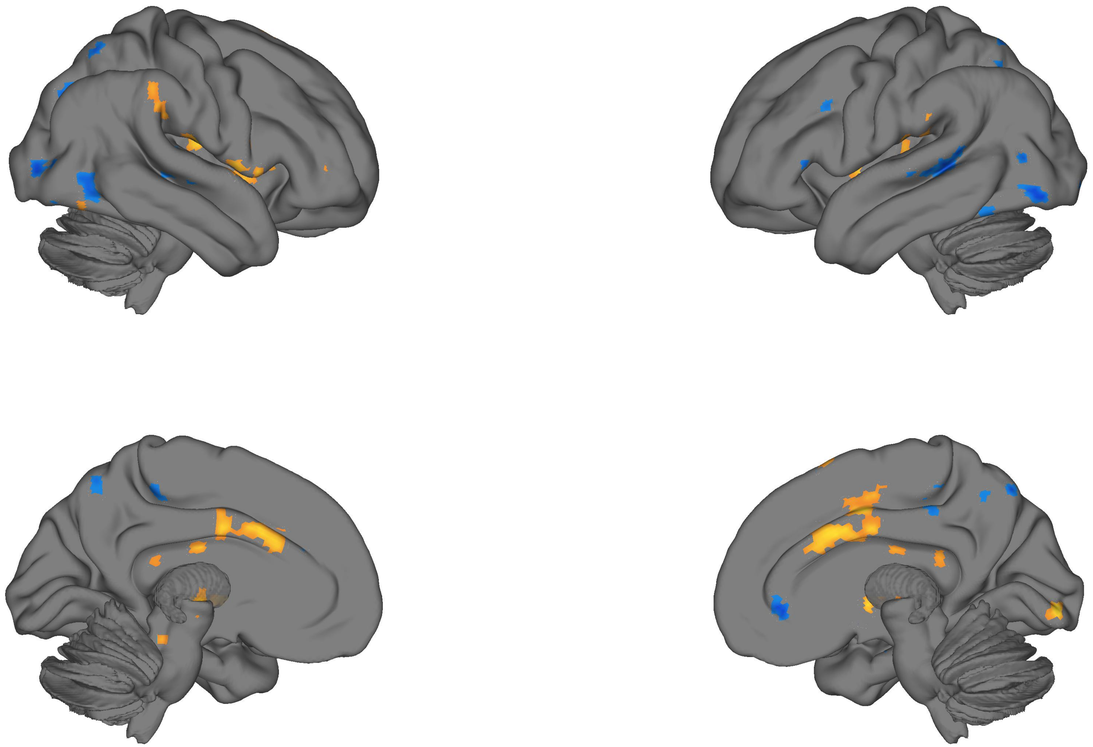
* These authors contributed equally as first authors.

** These authors contributed equally as last authors.

COI Disclosure:

What is a neural signature?

Distinct pattern of brain activity associated with specific tasks, emotions, or sensations.



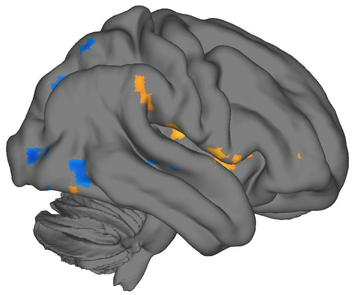
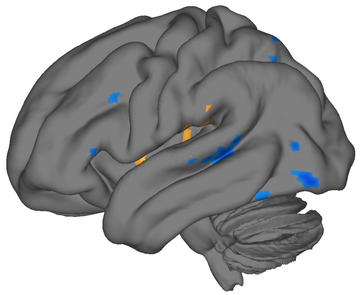
Example

Neurologic Pain Signature (NPS)

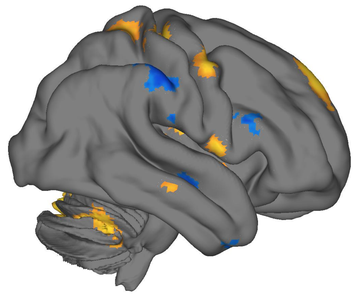
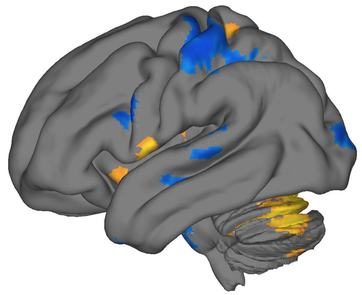
Neural pattern associated with physical pain intensity

Yellow = Positive pattern weights
Blue = Negative pattern weights

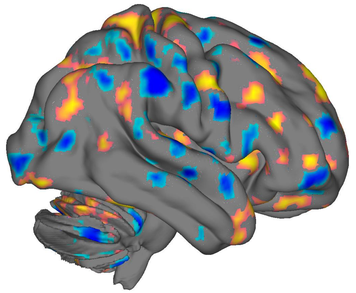
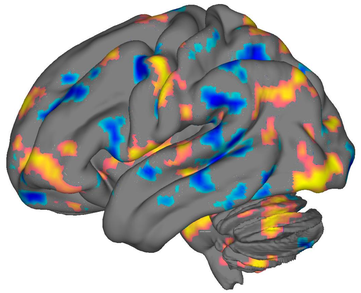
Four distinct neural patterns - a common factor?

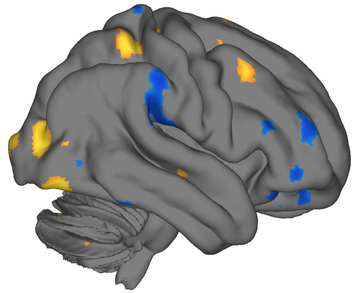
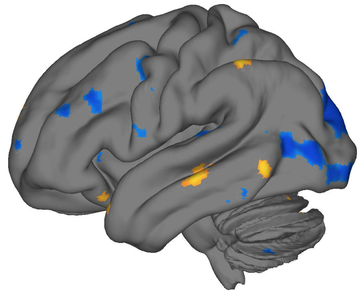
Pattern 1

Pattern 2

Pattern 3

Pattern 4

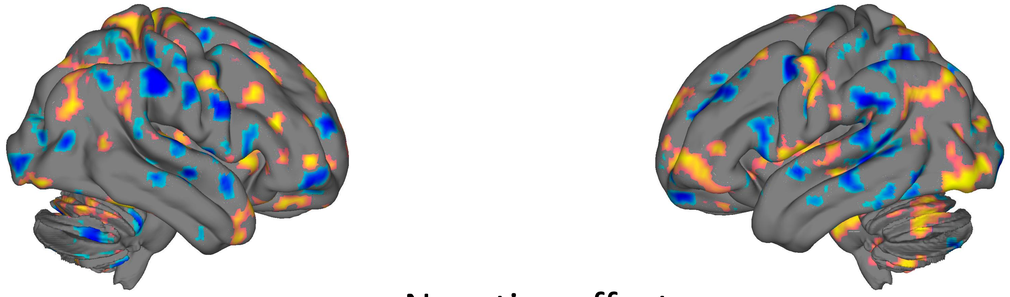
Four distinct neural patterns – all pain-related



NPS - physical pain intensity



SIIPS - Pain experience beyond stimulus intensity

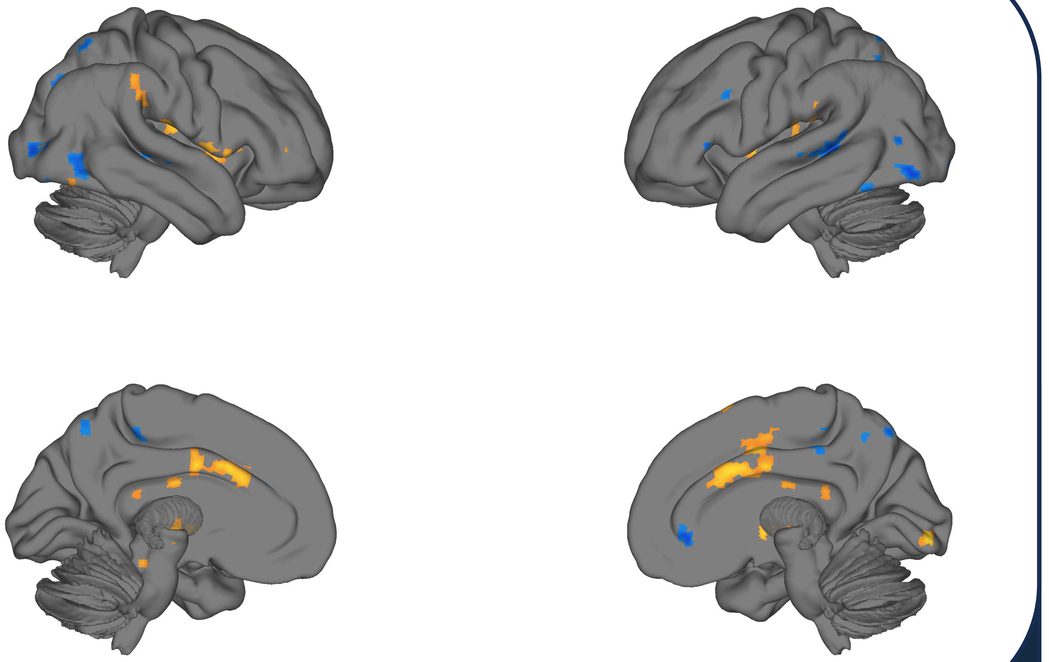


MPA2 General - Negative affect across different aversive experiences



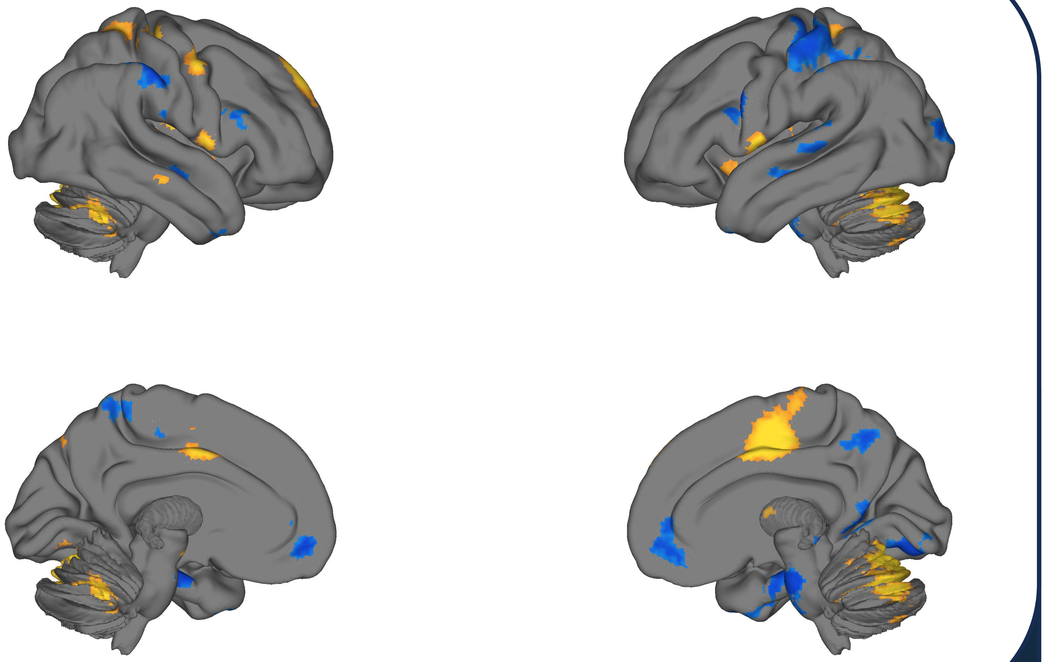
VPS – Pain observed in others

Two complementary signatures included in our analyses



Neurologic Pain Signature (NPS)

Distributed brain responses associated with acute, experimentally evoked physical pain.



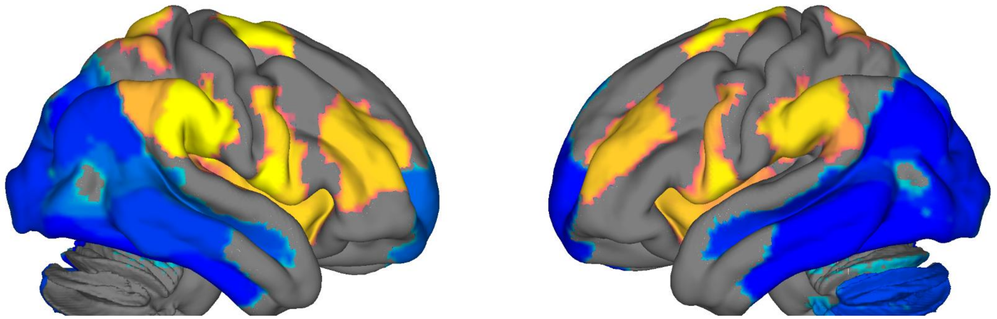
Stimulus Intensity-Independent Pain Signature (SIIPS)

Captures additional pain-related information not explained by stimulus intensity and the NPS.

Two complementary views of pain processing

01 Traditional BOLD GLM

Where do task-related brain responses occur?

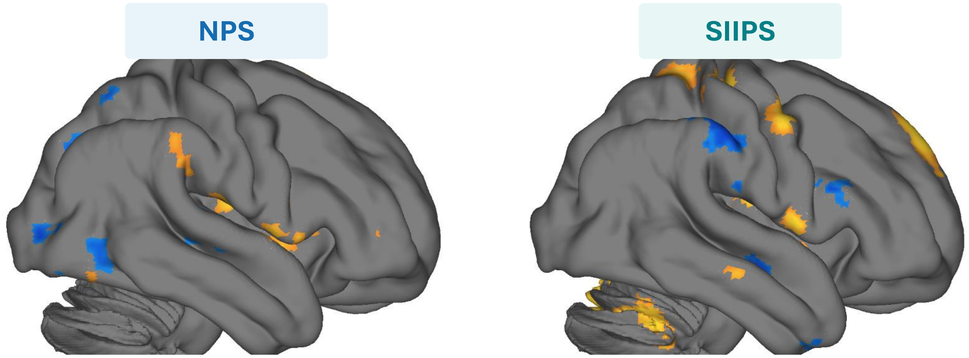


Example: high vs low inflation contrast

REGIONAL EFFECTS • Where and in which direction?

02 Multivariate neural signatures

How strongly are pain-related patterns expressed?



Predefined pattern weights

PATTERN EXPRESSION • How much of NPS / SIIPS?

THE CLBP QUESTION

Do these measures differ between people with CLBP and pain-free controls?

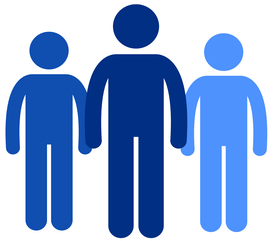
Secondary analyses performed on baseline dataset from

JAMA Psychiatry | Original Investigation

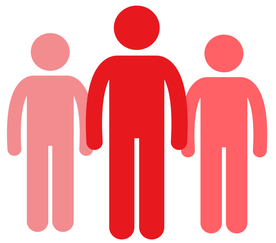
Effect of Pain Reprocessing Therapy vs Placebo and Usual Care for Patients With Chronic Back Pain
A Randomized Clinical Trial

Yoni K. Ashar, PhD; Alan Gordon, LCSW; Howard Schubiner, MD; Christie Uipi, LCSW; Karen Knight, MD; Zachary Anderson, BS; Judith Carlisle, MA; Laurie Polisky, BA; Stephan Geuter, PhD; Thomas F. Flood, MD, PhD; Philip A. Kragel, PhD; Sona Dimidjian, PhD; Mark A. Lumley, PhD; Tor D. Wager, PhD

Sample characteristics

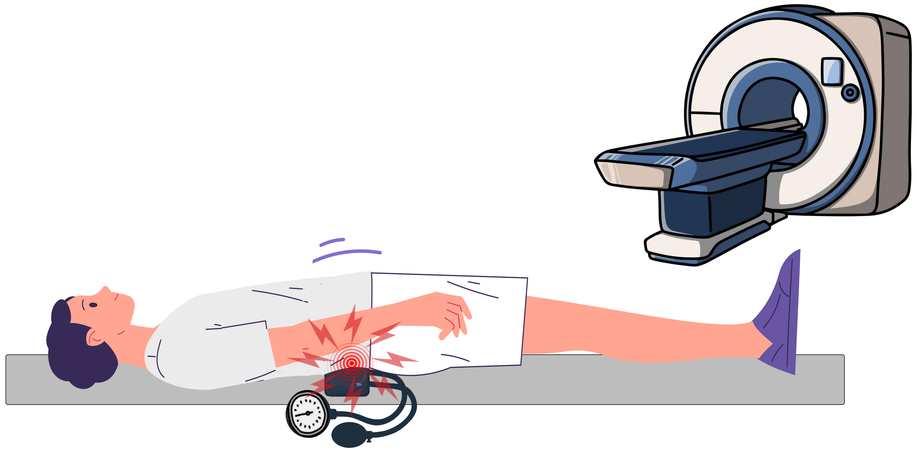


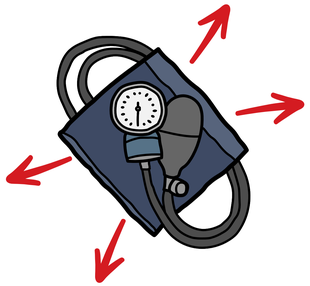
145 persons with CLBP
41.0 ± 15.5 years
54% female



50 pain-free controls
40.4 ± 14.3 years
54% female

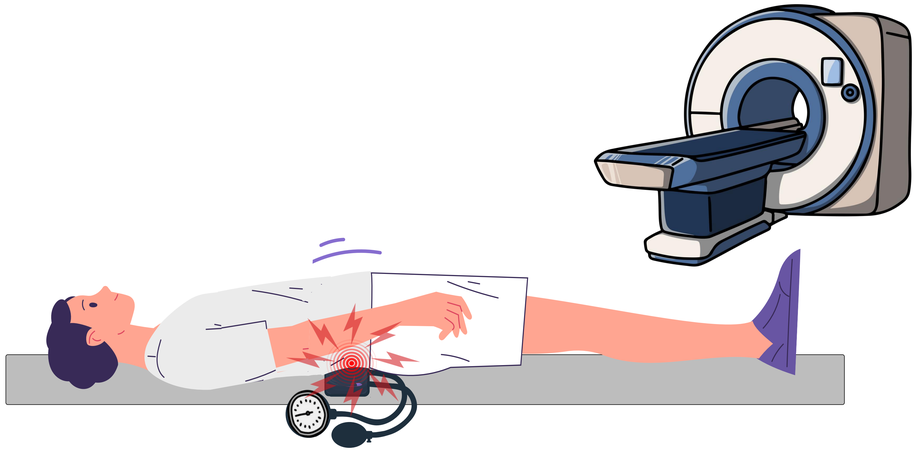
Functional magnetic resonance imaging: Evoked back pain task

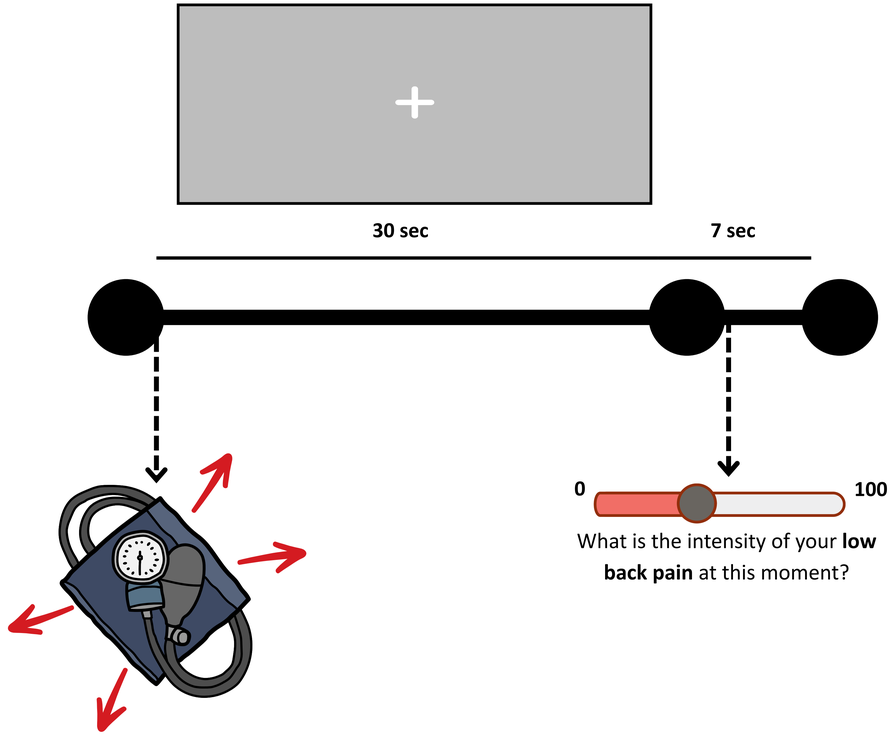




- Four inflation levels – 5 trials each
- Randomized order

Functional magnetic resonance imaging: Evoked back pain task





Two complementary views of the same fMRI data

01

Parcelwise whole-brain analysis

Location and direction of task-related responses

02

Multivariate signature analysis

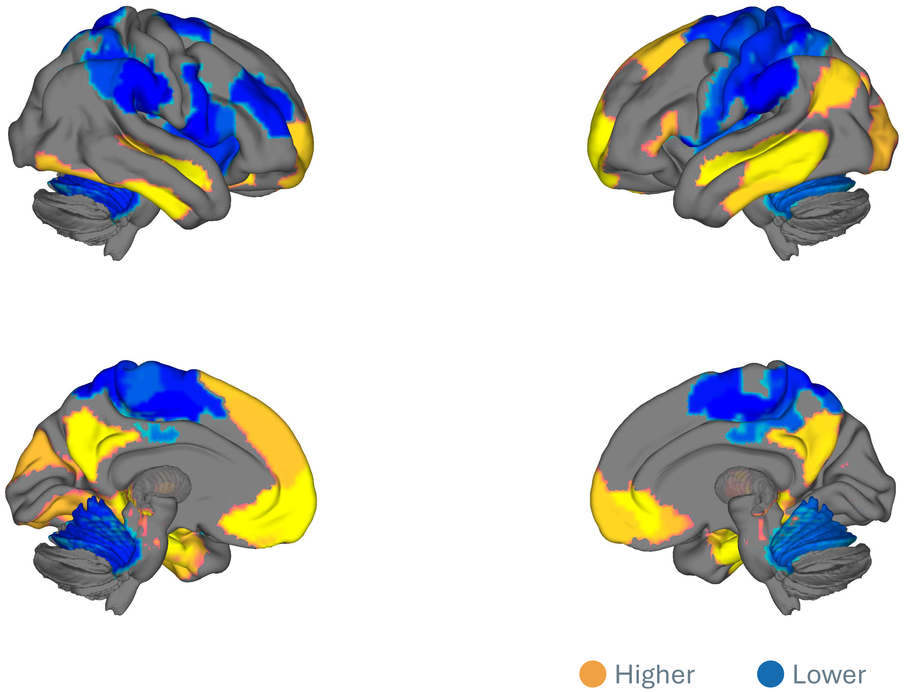
Expression of NPS and SIIPS patterns

Task responses are widespread and intensity dependent

Group-combined results · FDR-corrected q < 0.05

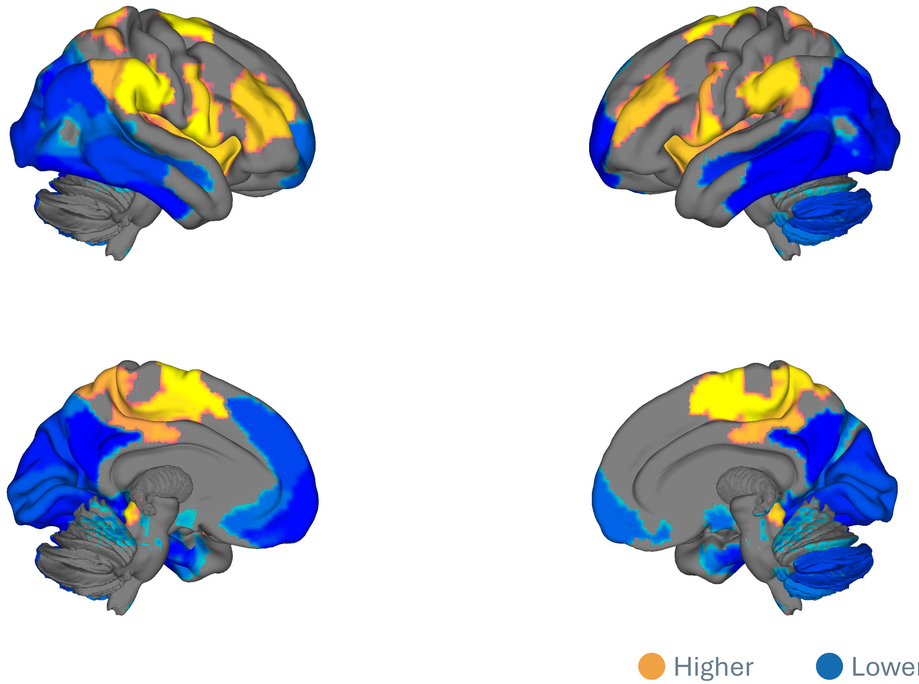
Task-evoked response

Inflation 2 vs implicit baseline



Intensity-dependent modulation

Inflation 4 vs inflation 2



No parcelwise group differences survived FDR

Despite robust task responses, significant CLBP–control differences did not emerge in individual parcels.

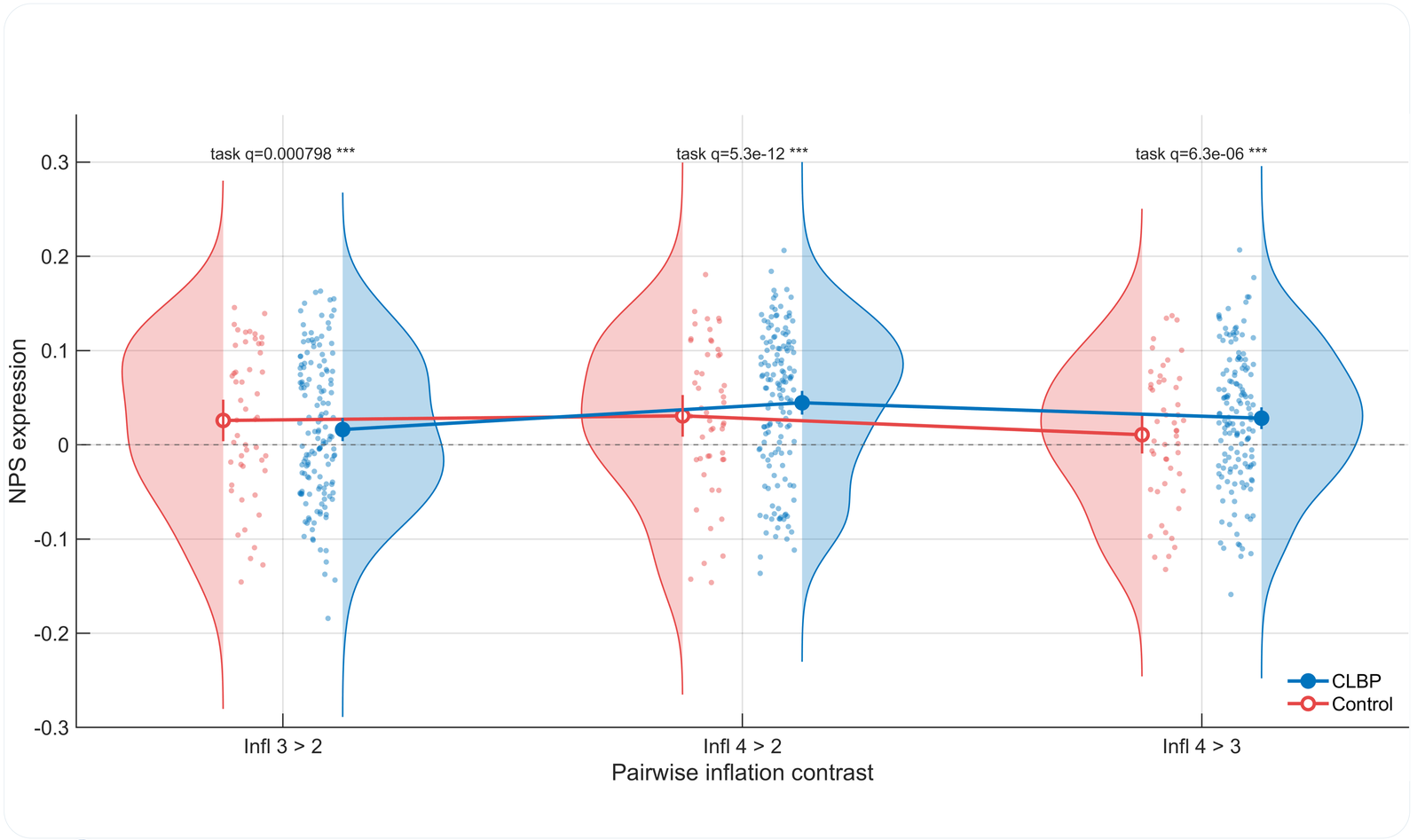
0

FDR-significant parcels for CLBP vs controls

But absence of regional differences does not imply identical distributed brain responses.

NEXT: TEST WHOLE-BRAIN SIGNATURE EXPRESSION

NPS expression differs across intensity contrasts



NPS

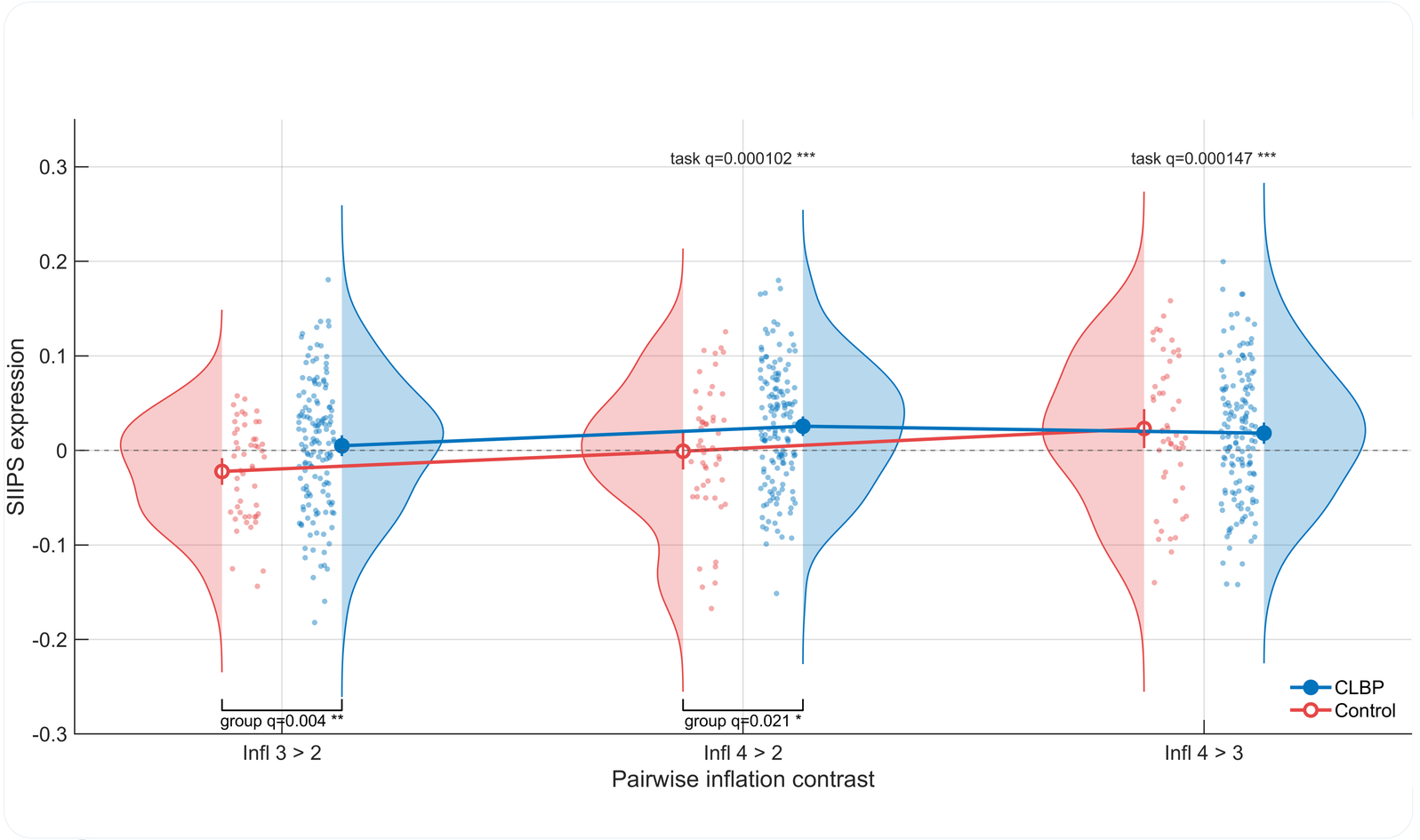
Task effects

q < 0.001

All three displayed pairwise contrasts

No significant group effects are annotated for NPS.

SIIPS expression is greater in CLBP for two contrasts



CLBP > CONTROLS

q = 0.004

Inflation 3 > 2

q = 0.021

Inflation 4 > 2

Storey FDR-corrected group effects

Task-related brain effects are not identical to group effects

01

Evoked back pain engages broad brain systems

Task and intensity effects appear across multiple regions.

02

Regional group effects were not detected

No parcelwise CLBP–control contrast survived FDR.

03

SIIPS reveals a group difference

CLBP showed greater expression for 3 > 2 and 4 > 2.